



A Three-Dimensional (3D) Semi-analytical Solution for the Ultimate End-Bearing Capacity of Rock-Socketed Shafts

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Received: 9 July 2021 / Accepted: 5 November 2021 / Published online: 22 November 2021
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Abstract

This study proposes a new semi-analytical solution for the ultimate end-bearing capacity of rock-socketed shafts with the consideration of three-dimensional (3D) strength and 3D geometry. The rock mass is assumed to be rigid-plastic and governed by a 3D Hoek–Brown (HB) criterion, and the method of characteristics is utilized to derive the governing equations in a cylindrical coordinate system. A Runge–Kutta-based iterative approach is adopted to solve the derived governing equations as an initial value problem via MATLAB. Eight test shafts with measured ultimate end-bearing capacity were analyzed to validate the proposed solution. Then, the proposed solution was compared with an analytical solution based on 2D HB criterion and an empirical factor for considering 3D effect, and a numerical simulation based on 2D HB criterion and 3D geometry to investigate the effects of 3D strength and 3D geometry. The results indicate that ignoring 3D strength and 3D geometry could significantly underestimate the ultimate end-bearing capacity of rock-socketed shafts. Finally, extensive parametric analyses were conducted with the proposed solution to explore the effects of rock mass properties, shaft dimensions, and the disturbance factor. The results show that the ultimate end-bearing capacity factor N_σ (which is the ratio of the ultimate end-bearing capacity q_u to the unconfined compressive strength of the intact rock σ_c) of rock-socketed shafts increases with rock mass constant m_i , geological strength index (GSI), the length of the shaft within soil layer H_s and the length of the shaft within rock layer H_r , while decreases with σ_c . Also, the disturbance factor D significantly affects the ultimate end-bearing capacity factor N_σ , especially for shafts in rock masses with a low GSI.

Keywords Rock-socketed shafts · End-bearing capacity · Method of characteristics · Disturbance · 3D strength · 3D geometry

1 Introduction

Rock-socketed shafts (also often called rock-socketed piles) are widely used to support different structures such as bridges, high-rise buildings, and wind turbines (Zhang 2004). The load from the shaft top is transferred to the surrounding ground via both sidewall and base, and the ultimate end-bearing capacity (also often called ultimate base resistance) q_u is especially important for a rock-socketed shaft when the side shear resistance is negligible or fully

mobilized at a small displacement. Traditionally, the q_u of rock-socketed shafts is often estimated using empirical formulas that relate the q_u with the unconfined compressive strength σ_c of intact rock. The different empirical relations between q_u and σ_c can be generally expressed as

$$q_u = \alpha \sigma_c^k, \quad (1)$$

where α and k are constants, which are determined based on the shaft test database (Zhang 2004). However, those empirical relations usually tend to overfit the shaft test database used, leading to low performance on designs at new sites. Also, the ground profile, shaft dimensions, and rock mass qualities are not considered by the empirical relations, although those factors can significantly affect the ultimate end-bearing capacity of rock-socketed shafts (EM 1994; ASSHTO 1996; CGS 2006).

Different theoretical solutions have been derived to take into account the effects of ground profile, shaft

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dimensions, and rock mass qualities. Serrano and Olalla (2002a, b) seemed to be the first to derive a theoretical solution of the ultimate end-bearing capacity of shafts in a rock mass governed by the Hoek–Brown (HB) criterion using the method of characteristics (Sokolovskii 1960, 1965). Using the same technique, the solution was further extended to shafts in rock masses governed by the modified HB criterion (Serrano et al. 2014) and the modified Mohr–Coulomb (MC) criterion (Galindo et al. 2017). More recently, the method of characteristics was further applied to shafts socketed in an inclined bedrock (Cai et al. 2020) and an anisotropic discontinuous rock mass with planes of weakness (Cao et al. 2021). However, the existing characteristics method-based solutions use a two-dimensional (2D) strength criterion and adopt an empirical factor (De Beer 1970) to empirically consider the three-dimensional (3D) geometry effect.

Besides the theoretical solutions based on the method of characteristics, Yang and Yin (2005) presented an upper bound solution to calculate the end-bearing capacity of shafts considering the modified HB criterion and the translational failure mechanism proposed by Michalowski (1997) and Soubra (1999). Yodsomjai et al. (2021) applied finite element limit analysis (FELA) to obtain an upper bound solution to the end-bearing capacity of shafts in rocks governed by the HB criterion. Gharsallaoui et al. (2020) used a spherical cavity expansion theory-based approach to analyze the end-bearing capacity of shafts in an HB criterion-governed rock mass following Yasufuku and Hyde (1995) and Vesic (1973). However, the assumption of an equivalent spherical cavity expansion process made by Gharsallaoui et al. (2020) was not verified by in-situ or laboratory tests.

In this study, a semi-analytical solution based on a 3D version of the HB criterion, the newly modified GZZ criterion (Chen et al. 2021a, b) is proposed to determine the ultimate end-bearing capacity of rock-socketed shafts using the method of characteristics. The weight of the rock mass and the 3D geometry of the cylindrical shaft are considered in deriving the governing equations and a Runge–Kutta-based iterative approach is adopted to solve the governing equations. Laboratory and in-situ test shafts with measured ultimate end-bearing capacity are analyzed to verify the proposed solution. The proposed solution is also compared with an existing analytical solution based on the 2D HB criterion and an empirical factor accounting for the 3D effect, and a numerical simulation based on the 2D HB criterion and 3D geometry to investigate the importance of 3D rock mass strength and 3D shaft geometry. Finally, extensive parametric studies are conducted with the present solution to evaluate the effects of rock mass properties, shaft dimensions, and the disturbance factor.

2 Failure Mechanism of Rock-Socketed Shaft and Rock Mass Failure Criterion

2.1 Basic Assumptions

Figure 1 shows the sketches of the failure surfaces of a shallowly socketed shaft (Fig. 1a) and a deeply socketed shaft (Fig. 1b). A central vertical load P is applied on the top of the shaft and causes the failure of the shaft in the two cases. The shaft has a diameter of B and a socket length of H_r , giving a socket ratio $n = H_r/B$. Once the stress under the shaft base reaches the ultimate bearing capacity q_u , large plastic deformation takes place along a failure surface and the shaft base fails. To differentiate the shallowly and deeply socketed shafts, a critical socket ratio n_L is defined as the ratio of the height of the failure surface above the base to the base diameter when

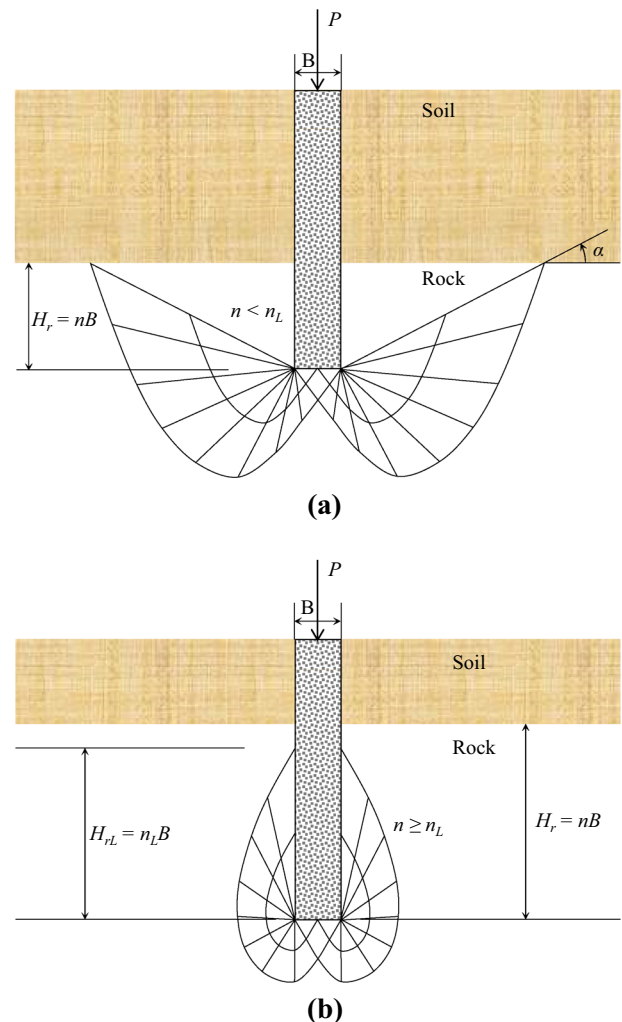


Fig. 1 Sketch of failure of (a) a shallowly socketed shaft; and (b) a deeply socketed shaft

the socket length is infinite. The shallowly socketed shaft ($n < n_l$) can induce a failure surface that intersects the interface of the soil and rock layers by an inclination angle α , while the deeply socketed shaft ($n \geq n_l$) is socketed so deeply that the failure surface can never reach the soil layer. The height of the failure surface above the base of the deeply socketed shaft is H_{rl} , which corresponds to the critical socket ratio n_L . Some basic assumptions are made as below:

- The rock mass is an isotropic rigid-plastic material, and the elastic deformation is negligible compared with the plastic deformation. And the strain-hardening/softening behavior of the rock mass is ignored.
- The rock mass is governed by a 3D version of the HB criterion, the newly modified GZZ criterion (Chen et al. 2021a, b). And the rock mass is weaker than the shaft concrete so that the failure would occur within the rock mass when the ultimate end-bearing capacity is reached.
- The intermediate principal stress involved in the adopted failure criterion is determined by considering the elastic circumferential strain as

$$\sigma_\theta = E\varepsilon_\theta^e + \mu(\sigma_r + \sigma_z) + \sigma_{\theta 0} - \mu(\sigma_{r0} + \sigma_{z0}), \quad (2a)$$

$$\varepsilon_\theta^e = -\frac{u_r}{r}, \quad (2b)$$

where σ_r , σ_θ , and σ_z are the effective stresses along the radial, circumferential and vertical directions, respectively; E and μ are Young's modulus and Poisson's ratio of the rock mass, respectively; σ_{r0} , $\sigma_{\theta 0}$, and σ_{z0} are the initial values of σ_r , σ_θ , and σ_z before the installation of the shaft, respectively; ε_θ^e is the elastic circumferential strain; u_r denotes the radial displacement; and r is the radial coordinate. According to research on circular shafts in soils (Lau 1988; Bolton and Lau 1993), the effect of σ_θ is to increase the bearing capacity. For conservative consideration, ε_θ^e is taken as zero and

$$\sigma_\theta = \mu(\sigma_r + \sigma_z) + \sigma_{\theta 0} - \mu(\sigma_{r0} + \sigma_{z0}). \quad (3)$$

- The characteristic lines for the stress field and the velocity field are the same. Hence, the shape of the failure surface is controlled by the instant friction angle of the newly modified GZZ criterion and is solved by the method of characteristics (Sokolovskii 1960, 1965). See details in Sect. 3.1.
- The unit weight of the rock mass γ_r is taken as 25 kN/m³ for simplification. The top of the rock mass is flat, and the shaft is vertically installed.
- The base of the shaft is smooth and thus the stress on the shaft base is in the vertical direction.

- The side of the shaft is smooth and thus only the normal stress on the shaft sidewall in Fig. 1b is involved in the boundary condition.

2.2 Newly Modified GZZ Criterion

The ultimate end-bearing capacity of a single rock-socketed shaft is an axisymmetric problem with the circumferential stress involved in the radial equilibrium equation. Therefore, a 3D failure criterion rather than a 2D failure criterion should be used. Considering the wide application of the HB criterion in rock engineering, a 3D version of the HB criterion, the newly modified GZZ criterion (Chen et al. 2021a, b), is adopted and its expression is given as

$$f = \left[\left(\frac{q}{\sigma_c} \right)^{2/a-2} + m_b \left(\frac{q}{\sigma_c} \right)^{1/a-1} \right] (I_1^* I_2^* - 9I_3^*) - 2m_b^2 I_3^* = 0, \quad (4)$$

where

$$q = \sqrt{I_1^* - 3I_2^*}, \quad (5a)$$

$$I_2^* = \sigma_1^* \sigma_2^* + \sigma_2^* \sigma_3^* + \sigma_3^* \sigma_1^*, \quad (5b)$$

$$I_1^* = 3p^* = \sigma_1^* + \sigma_2^* + \sigma_3^*, \quad (5c)$$

$$I_3^* = \sigma_1^* \sigma_2^* \sigma_3^*, \quad (5d)$$

$$\sigma_i^* = \sigma_i + \frac{\sigma_c}{m_b} s, \quad i = 1, 2, 3, \quad (5e)$$

$$m_b = \exp \left(\frac{\text{GSI}-100}{28-14D} \right) m_i, \quad (5f)$$

$$s = \exp \left(\frac{\text{GSI}-100}{9-3D} \right), \quad (5g)$$

$$a = 0.5 + \frac{1}{6} \left(e^{-\frac{\text{GSI}}{15}} - e^{-\frac{20}{3}} \right), \quad (5h)$$

in which σ_1 , σ_2 , and σ_3 are the major, intermediate and minor effective principal stresses, respectively; m_i is a constant for the intact rock; m_b , a and s are parameters of the rock mass; GSI means geology strength index; and D is a disturbance factor representing the effect of construction on the rock mass.

It should be noted that σ_θ is the intermediate effective principal stress based on the axisymmetric condition of a shaft under a central vertical load. Therefore, Eq. (3) can be rewritten as

$$\begin{aligned}\sigma_2 = \sigma_\theta &= \mu(\sigma_r + \sigma_z) + \sigma_{\theta 0} - \mu(\sigma_{r0} + \sigma_{z0}) \\ &= \mu(\sigma_1 + \sigma_3) + \sigma_{\theta 0} - \mu(\sigma_{r0} + \sigma_{z0}).\end{aligned}\quad (6)$$

Substituting Eq. (6) into Eq. (4), the strength criterion $f = 0$ can be rewritten as a function of σ_1 and σ_3 . Adopting the two Lamé parameters, $p_L = (\sigma_1 + \sigma_3)/2$ and $q_L = (\sigma_1 - \sigma_3)/2$, the newly modified GZZ criterion can be expressed in the p_L – q_L plane as shown in Fig. 2.

According to the geometric relation in Fig. 2, the relationship between (p_L, q_L) and (σ, τ) can be expressed as

$$\tau = q_L \cos \rho, \quad (7a)$$

$$\sigma = p_L - q_L \sin \rho, \quad (7b)$$

$$\tan \rho = \frac{d\tau}{d\sigma}, \quad (7c)$$

where ρ is the instant tangential friction angle.

Performing differentiation to Eqs. (7a) and (7b), ρ can be related to p_L and q_L as

$$\sin \rho = \frac{dq_L}{dp_L}. \quad (8)$$

Alternatively, ρ can be expressed with respect to σ_1 and σ_3 as

$$\sin \rho = \frac{dq_L}{dp_L} = \frac{d\sigma_1 - d\sigma_3}{d\sigma_1 + d\sigma_3} = \frac{d\sigma_1/d\sigma_3 - 1}{d\sigma_1/d\sigma_3 + 1}. \quad (9)$$

3 Solution to End-Bearing Capacity of Rock-Socketed Shafts

3.1 Method of Characteristics

Figure 3 shows the characteristic lines of the newly modified GZZ criterion-based rock mass under the axisymmetric assumption in the cylindrical coordinate system. The differential equations of the two characteristic lines are (Serrano and Olalla 2002a)

$$\frac{dr}{dz} = \tan(\psi \pm \omega), \quad (10)$$

where r and z are the radial and vertical coordinates, respectively; ψ denotes the inclination of the major effective principal stress relative to the positive axis z ; and ω denotes the angle between the direction of the major effective principal stress and the two characteristic lines and can be expressed as

$$\omega = \frac{\pi}{4} - \frac{\rho}{2}. \quad (11)$$

Considering the axisymmetric condition, the equilibrium equations in the cylindrical coordinate system can be simplified as

$$\frac{\partial \sigma_r}{\partial r} + \frac{\partial \tau_{rz}}{\partial z} + \frac{\sigma_r - \sigma_\theta}{r} = 0, \quad (12a)$$

Fig. 2 Newly modified GZZ criterion expressed in p_L – q_L plane

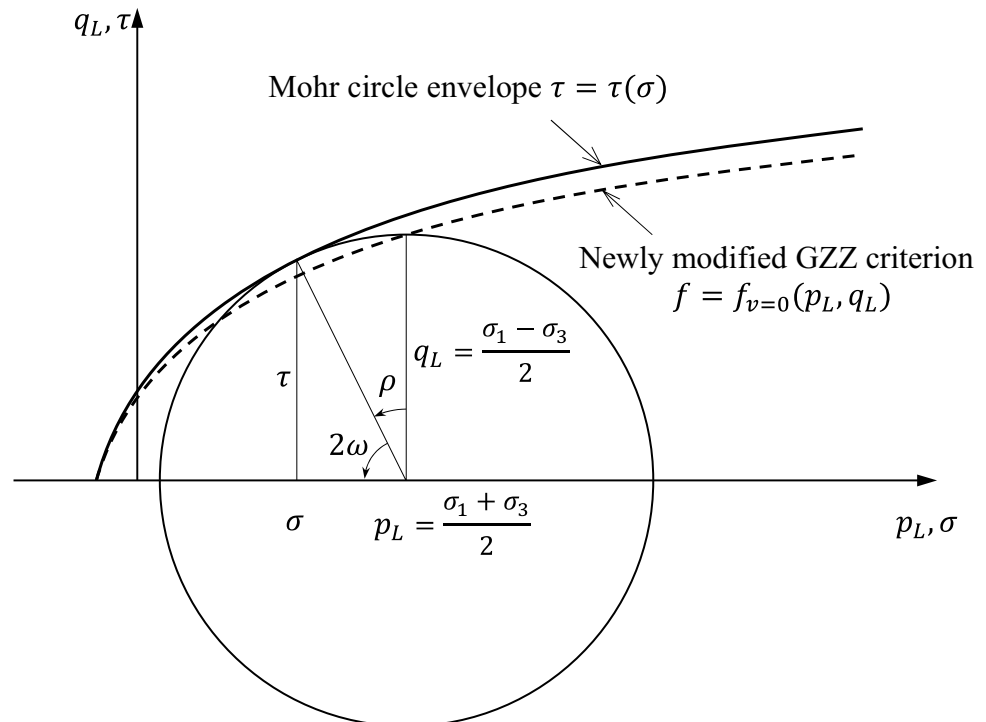
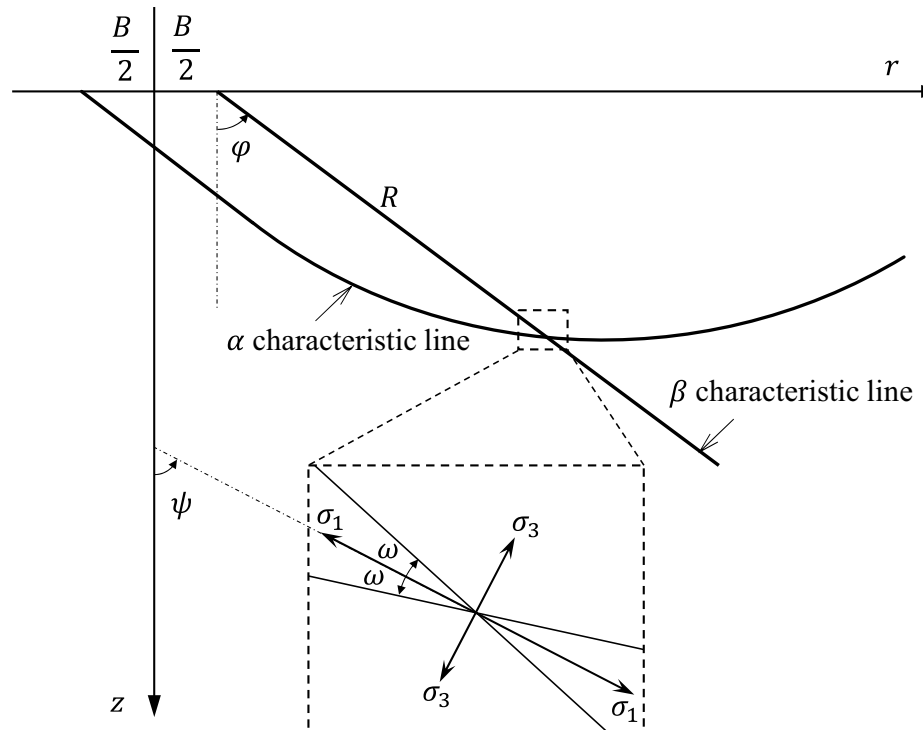


Fig. 3 Characteristic lines in a cylindrical coordinate system



$$\frac{\partial \tau_{rz}}{\partial r} + \frac{\partial \sigma_z}{\partial z} + \frac{\tau_{rz}}{r} = \gamma_r, \quad (12b)$$

where γ_r is the unit weight of the rock mass. It should be noted that the previous studies (Serrano and Olalla 2002a, b; Serrano et al. 2014; Galindo et al. 2017; Cai et al. 2020; Cao et al. 2021) assumed a plane strain condition and ignored the weight of rock mass.

The three stresses within the rz plane can be obtained based on Fig. 3 and considering stress rotation as

$$\sigma_r = p_L - q_L \cos 2\psi, \quad (13a)$$

$$\sigma_z = p_L + q_L \cos 2\psi, \quad (13b)$$

$$\tau_{rz} = q_L \sin 2\psi. \quad (13c)$$

Combining Eqs. (6), (12a, 12b), and (13a–13c), the following governing equations along the two characteristic lines can be obtained

$$\begin{aligned} & \cos \rho dp_L \pm 2q_L d\psi + \frac{(p_L - \sigma_\theta) \cos \rho - q_L \cos(2\psi \mp \rho)}{2r} dr \\ & + \frac{\pm(p_L - \sigma_\theta) \sin \rho + q_L \sin(2\psi \mp \rho)}{2r} dz = \frac{\gamma_r}{2} \cos \rho dz \mp \frac{\gamma_r}{2} \sin \rho dr. \end{aligned} \quad (14)$$

As for the α characteristic line, its governing equations, i.e., Eqs. (10) and (14), can be given as

$$\begin{aligned} & \cos \rho dp_L + 2q_L d\psi + \frac{(p_L - \sigma_\theta) \cos \rho - q_L \cos(2\psi - \rho)}{2r} dr \\ & + \frac{(p_L - \sigma_\theta) \sin \rho + q_L \sin(2\psi - \rho)}{2r} dz = \frac{\gamma_r}{2} \cos \rho dz - \frac{\gamma_r}{2} \sin \rho dr, \end{aligned} \quad (15a)$$

$$\frac{dr}{dz} = \tan(\psi + \omega). \quad (15b)$$

And in the polar coordinates (R, θ, φ) as shown in Fig. 3, the r , z and ψ of the α characteristic line can be expressed as

$$r = R \sin \varphi + \frac{B}{2}, \quad (16a)$$

$$z = R \cos \varphi, \quad (16b)$$

$$\psi = \varphi + \omega = \varphi - \frac{\rho}{2} + \frac{\pi}{4}. \quad (16c)$$

Performing differentiation to Eqs. (16a–16c)

$$dr = \sin \varphi dR + R \cos \varphi d\varphi, \quad (17a)$$

$$dz = \cos \varphi dR - R \sin \varphi d\varphi. \quad (17b)$$

And Eq. (15b) can be expressed in the polar coordinates with the help of Eqs. (17a, 17b) as

$$\frac{dR}{d\varphi} = R \tan \rho. \quad (18)$$

Substituting Eqs. (17a, 17b) and (18) into Eqs. (15a, 15b), one has the governing equation with respect to p_L , ψ and φ as

$$\cos \rho dp_L + 2q_L d\psi + A_\varphi d\varphi = 0, \quad (19)$$

where

$$A_\varphi = \left[\frac{(p_L - \sigma_\theta) \cos \rho - q_L \cos(2\psi - \rho)}{2R \sin \varphi + B} + \frac{\gamma_c}{2} \sin \rho \right] R(\sin \varphi \tan \rho + \cos \varphi) + \left[\frac{(p_L - \sigma_\theta) \sin \rho + q_L \sin(2\psi - \rho)}{2R \sin \varphi + B} - \frac{\gamma_c}{2} \cos \rho \right] R(\cos \varphi \tan \rho - \sin \varphi). \quad (20)$$

Note that φ is a function of ψ and ρ as indicated by Eq. (16c). Combining Eq. (16c) with Eq. (19) and taking ρ as a function of p_L according to Eq. (8), Eq. (19) is further simplified as

$$\begin{aligned} \cos \rho dp_L + 2q_L d\psi + A_\varphi d\psi + \frac{A_\varphi}{2} d\rho \\ = \left(\cos \rho + \frac{A_\varphi}{2} \frac{\partial \rho}{\partial p_L} \right) dp_L + (2q_L + A_\varphi) d\psi = 0, \end{aligned} \quad (21)$$

or

$$\frac{dp_L}{d\psi} = - \frac{2q_L + A_\varphi}{\cos \rho + \frac{A_\varphi}{2} \frac{\partial \rho}{\partial p_L}}. \quad (22)$$

Also, the newly modified GZZ criterion and Eq. (6) should be satisfied along the α characteristic line; hence

$$df = \frac{\partial f}{\partial \sigma_1^*} d\sigma_1 + \frac{\partial f}{\partial \sigma_2^*} d\sigma_2 + \frac{\partial f}{\partial \sigma_3^*} d\sigma_3 = 0, \quad (23a)$$

$$d\sigma_\theta = d\sigma_2 = \mu(d\sigma_r + d\sigma_z) = \mu(d\sigma_1 + d\sigma_3) = 2\mu dp_L. \quad (23b)$$

Combining Eqs. (23a) and (23b) yields

$$d\sigma_1 = \frac{2}{1+A_{13}} dp_L, \quad (24a)$$

$$d\sigma_2 = \frac{2A_{12}}{1+A_{13}} dp_L = 2\mu dp_L, \quad (24b)$$

$$d\sigma_3 = \frac{2A_{13}}{1+A_{13}} dp_L, \quad (24c)$$

$$A_{12} = -\mu \frac{\frac{\partial f}{\partial \sigma_1^*} - \frac{\partial f}{\partial \sigma_3^*}}{\frac{\partial f}{\partial \sigma_3^*} + \mu \frac{\partial f}{\partial \sigma_2^*}}, \quad (24d)$$

$$A_{13} = -\frac{\frac{\partial f}{\partial \sigma_1^*} + \mu \frac{\partial f}{\partial \sigma_2^*}}{\frac{\partial f}{\partial \sigma_3^*} + \mu \frac{\partial f}{\partial \sigma_2^*}}. \quad (24e)$$

The details of Eqs. (24a–24e) are given in Appendix A1. Substituting Eq. (24e) into Eq. (9) yields

$$\sin \rho = \frac{d\sigma_1/d\sigma_3 - 1}{d\sigma_1/d\sigma_3 + 1} = \frac{1-A_{13}}{A_{13}+1}, \quad (25a)$$

$$\cos \rho = \sqrt{1 - \sin^2 \rho} = \sqrt{1 - \left(\frac{d\sigma_1/d\sigma_3 - 1}{d\sigma_1/d\sigma_3 + 1} \right)^2} = \sqrt{1 - \left(\frac{1-A_{13}}{A_{13}+1} \right)^2}. \quad (25b)$$

Performing differentiation to Eq. (25a), one has

$$\frac{d\rho}{dp_L} = - \frac{2}{\cos \rho (A_{13}+1)^2} \frac{dA_{13}}{dp_L}, \quad (26a)$$

$$\frac{dA_{13}}{dp_L} = \frac{2}{1+A_{13}} \frac{dA_{13}}{d\sigma_1^*} + \frac{2A_{12}}{1+A_{13}} \frac{dA_{13}}{d\sigma_2^*} + \frac{2A_{13}}{1+A_{13}} \frac{dA_{13}}{d\sigma_3^*}. \quad (26b)$$

The details of Eq. (26b) are provided in Appendix A2.

Substituting Eq. (22) into Eq. (24a–24e)

$$\frac{d\sigma_1}{d\psi} = - \frac{2}{1+A_{13}} \frac{(\sigma_1 - \sigma_3 + A_\varphi)}{\cos \rho + \frac{A_\varphi}{2} \frac{\partial \rho}{\partial p_L}}, \quad (27a)$$

$$\frac{d\sigma_2}{d\psi} = - \frac{2A_{12}}{1+A_{13}} \frac{(\sigma_1 - \sigma_3 + A_\varphi)}{\cos \rho + \frac{A_\varphi}{2} \frac{\partial \rho}{\partial p_L}}, \quad (27b)$$

$$\frac{d\sigma_3}{d\psi} = - \frac{2A_{13}}{1+A_{13}} \frac{(\sigma_1 - \sigma_3 + A_\varphi)}{\cos \rho + \frac{A_\varphi}{2} \frac{\partial \rho}{\partial p_L}}. \quad (27c)$$

Similarly, combining Eqs. (18) and (26a, 26b), the instant friction angle ρ and the coordinates of the α characteristic line can be determined

$$\frac{d\rho}{d\psi} = \frac{\partial \rho}{\partial p_L} \frac{dp_L}{d\psi} = - \frac{(\sigma_1 - \sigma_3 + A_\varphi)}{\cos \rho + \frac{A_\varphi}{2} \frac{\partial \rho}{\partial p_L}} \frac{\partial \rho}{\partial p_L}, \quad (28a)$$

$$\frac{d\varphi}{d\psi} = 1 + \frac{1}{2} \frac{d\rho}{d\psi} = 1 - \frac{(\sigma_1 - \sigma_3 + A_\varphi)}{2 \cos \rho + A_\varphi \frac{\partial \rho}{\partial p_L}} \frac{\partial \rho}{\partial p_L}, \quad (28b)$$

$$\frac{dR}{d\psi} = \frac{dR}{d\varphi} \frac{d\varphi}{d\psi} = \left[1 - \frac{(\sigma_1 - \sigma_3 + A_\varphi)}{2 \cos \rho + A_\varphi \frac{\partial \rho}{\partial p_L}} \frac{\partial \rho}{\partial p_L} \right] R \tan \rho. \quad (28c)$$

It should be noted that Eqs. (27a–27c) and (28a–28c) are the governing equations along the α characteristic line (failure surface) with ψ as the basic variable. To determinate the stresses along the characteristic line, a Runge–Kutta-based iterative approach is proposed here to numerically integrate Eqs. (27a–27c) from boundary 1 to boundary 2 as shown in Fig. 4. Similarly, the shape of the characteristic line can be determined by integrating Eqs. (28a–28c). Therefore, the next step is to determine the initial values at boundary 1 and ψ at boundary 2.

3.2 Determination of Boundary Conditions

Figure 4 shows the two boundary conditions involved in the present solution: boundary 1 is the final extension of the failure surface and boundary 2 denotes the shaft base following

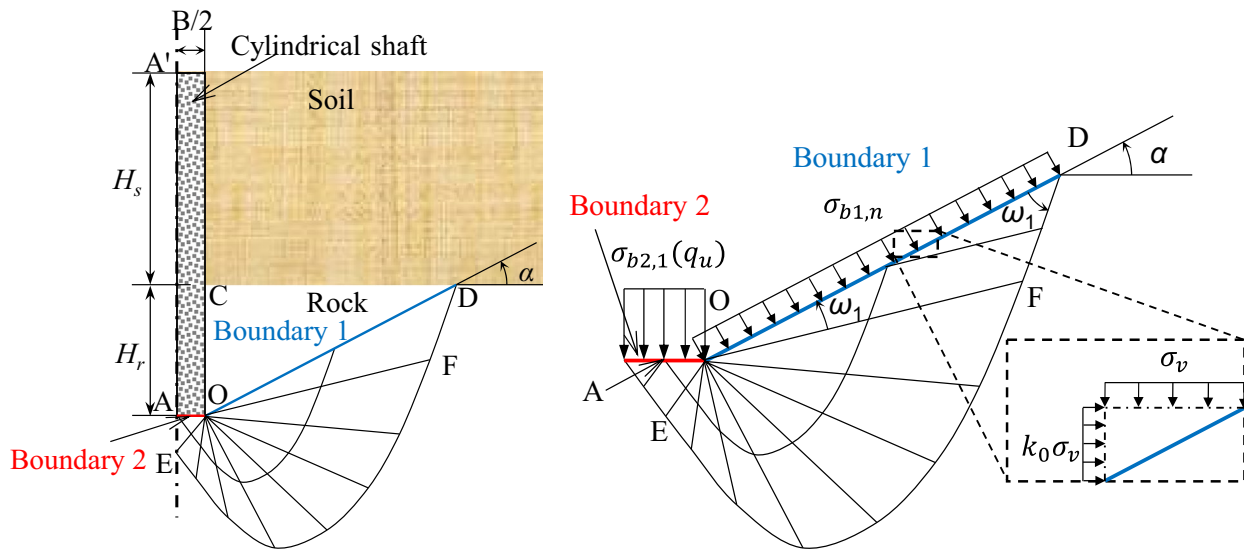


Fig. 4 Boundary conditions

Serrano and Olalla (2002a). Once the failure surface (EFD in Fig. 4) forms due to the load on the shaft head, the uniform ultimate bearing capacity $q_u (= \sigma_{b2,1})$, major effective principal stress at boundary 2) is normally applied on the shaft base (boundary 2), and an exterior cone surface (boundary 1) ascends with an angle α under the overburden stress. Especially, $\alpha = 90^\circ$ for a deeply socketed shaft. To be simple, the distributed stresses on boundary 1 and boundary 2 are assumed to be uniform. A rock-socketed shaft under a central vertical load should be considered under an axisymmetric condition rather than a plane strain condition assumed by many previous researchers (Serrano and Olalla 2002a; Serrano et al. 2014; Galindo et al. 2017). Hence, the equivalent uniform stress on boundary 1 should be determined by considering the 3D geometry of the shaft, as discussed in detail in Sect. 3.2.1 below. Furthermore, both the stress states at boundaries 1 and 2 should be on the newly modified GZZ criterion surface as given by Eq. (4) and Fig. 2.

3.2.1 Boundary 1

As shown in Fig. 4, a cylindrical shaft, which is axisymmetric about AA', is penetrated through the soil layer and socketed in the rock layer. The total vertical load on boundary 1

should be the weight of OCD and the upper soil layer. Considering the axisymmetric nature of the shaft, block OCD has a ring shape in 3D space and its weight is

$$W_r = \pi \gamma_r \left(\frac{B}{2} + \frac{H_r}{3} \cot \alpha \right) H_r^2 \cot \alpha, 0^\circ \leq \alpha < 90^\circ. \quad (29)$$

Therefore, the average vertical stress on boundary 1 caused by the weight of rock can be determined

$$\sigma_{v,r} = \frac{W_r}{A} = \frac{\pi \gamma_r \left(\frac{B}{2} + \frac{H_r}{3} \cot \alpha \right) H_r^2 \cot \alpha}{\pi (B + H_r \cot \alpha) H_r \cot \alpha} = \frac{3B + 2H_r \cot \alpha}{6(B + H_r \cot \alpha)} \gamma_r H_r, 0^\circ \leq \alpha < 90^\circ. \quad (30)$$

As for the upper soil layer

$$\sigma_{v,s} = \gamma_s H_s. \quad (31)$$

So, the average overburden exerted on OC (boundary 1) is

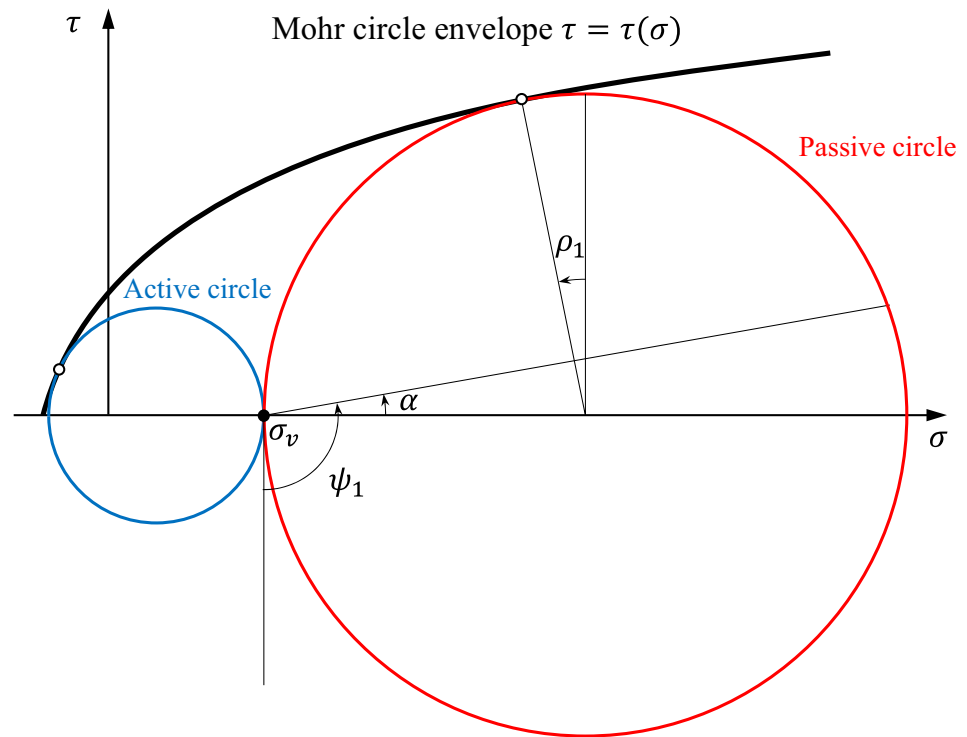
$$\sigma_v = \sigma_{v,r} + \sigma_{v,s} = \frac{3 + 2n \cot \alpha}{6(1 + n \cot \alpha)} \gamma_r B n + \gamma_s H_s, 0^\circ \leq \alpha < 90^\circ. \quad (32)$$

As for a deeply socketed shaft ($\alpha = 90^\circ$ and $n > n_L$)

$$\begin{aligned} \sigma_v &= \sigma_{v,r} + \sigma_{v,s} = \gamma_r (H_r - H_{rL}/2) + \gamma_s H_s \\ &= \gamma_r B (n - n_L/2) + \gamma_s H_s, \alpha = 90^\circ. \end{aligned} \quad (33)$$

To be simple, the coefficient of lateral earth pressure is taken as $k_0 = 1$. Hence, the normal stress $\sigma_{b1,n}$ and shear stress τ_{b1} on boundary 1 can be determined as

Fig. 5 Active and passive Mohr circles for boundary 1



$$\sigma_{b1,n} = \sigma_v, \quad (34a)$$

$$\tau_{b1} = 0. \quad (34b)$$

Once the failure surface is completely formed, a counterclockwise rotational movement around point O, i.e., the velocity field, is generated and each point within that velocity field fails. As for the points on boundary 1, two Mohr circles can be plotted tangent to the strength curve: the active circle and the passive circle (Fig. 5). Note that the passive circle corresponds to the counterclockwise rotation kinematics (Serrano and Olalla 2002a). Under this context, the normal stress on boundary 1 should be the minor effective principal stress and thus the direction of σ_1 should be along boundary 1 based on the passive moving mode. Hence, the direction of σ_1 is

$$\psi_1 = \frac{\pi}{2} + \alpha. \quad (35)$$

The value of $\sigma_{b1,n}$ depends on α and the shaft dimensions as shown in Eqs. (32)–(34). The determination of α can be achieved by an iterative approach and will be discussed in Sect. 3.4. Once $\sigma_{b1,n}$ is determined, it can be substituted into the newly modified GZZ criterion and Eq. (3) so that all the three principal stresses at boundary 1 are determined. And

ρ_1 can be determined from Eqs. (4) and (9), ψ_1 from Eq. (35), and R_1 from the equations below as a function of n and α

$$R_1 = n \frac{B}{\sin \alpha}, 0 \leq \alpha < \frac{\pi}{2} \text{ for shallowly socketed shaft}, \quad (36a)$$

$$R_1 = n_L B, \alpha = \frac{\pi}{2} \text{ for deeply socketed shaft}. \quad (36b)$$

It should be noted that both the vertical and horizontal initial stresses are considered on boundary 1 as indicated by Eqs. (34a, 34b) in the current study, while only the vertical stress is considered in the previous studies such as that by Serrano et al. (2014). Therefore, different from Serrano et al. (2014), there is no difference between small and great overburden pressures, and thus only two failure mechanisms are considered in the proposed solution.

3.2.2 Boundary 2

As for boundary 2, it is assumed there is no friction on the base of the shaft. Hence, the major effective principal stress is in the vertical direction, which indicates

$$\psi_2 = 0. \quad (37)$$

And the stresses should be determined by solving the governing equations.

3.3 Socket Ratio

As discussed in Sect. 2.1, the shallowly and deeply socketed shafts are classified based on the socket ratio. The socket ratio n as shown in Fig. 4 can be rewritten as (Serrano and Olalla 2002a)

$$n = \frac{H_r}{B} = \frac{OD \sin \alpha}{2OA} = \frac{\sin \alpha}{2} \left(\frac{OD}{OF} \right) \left(\frac{OF}{OE} \right) \left(\frac{OE}{OA} \right). \quad (38)$$

According to the geometry as shown in Fig. 4

$$\frac{OD}{OF} = \frac{\sin(\pi - 2\omega_1)}{\sin \omega_1} = 2 \cos \omega_1, \quad (39a)$$

$$\frac{OE}{OA} = \frac{1}{\sin \omega_2}. \quad (39b)$$

Integrating Eq. (18) from point E to point F

$$\frac{OF}{OE} = \exp \left(\int_{\varphi_E}^{\varphi_F} \tan \rho d\varphi \right), \quad (40)$$

where φ_E and φ_F are the angular coordinates of E and F , respectively.

Therefore

$$n = \sin \alpha \frac{\cos \omega_1}{\sin \omega_2} \exp \left(\int_{\varphi_E}^{\varphi_F} \tan \rho d\varphi \right). \quad (41)$$

The angle α increases as n increases and $\alpha = 90^\circ$ as n approaches n_L . Under this context, for a deeply socketed shaft with $n \geq n_L$, the plastic bulb develops to the end where it closes upon the shaft (Fig. 1b). Considering $\alpha = 90^\circ$ when $n \geq n_L$, Eq. (41) can be further simplified as

$$n_L = \frac{\cos \omega_1}{\sin \omega_2} \exp \left(\int_{\varphi_E}^{\varphi_F} \tan \rho d\varphi \right). \quad (42)$$

3.4 Solving Procedure

As discussed in Sect. 3.2.1, α and σ_v should be determined using an iterative approach. Furthermore, the critical socket ratio n_L should be iteratively determined as Eq. (42) changes with different overburdens. Therefore, the iterative solving procedure as shown in Fig. 6 is followed for the proposed semi-analytical solution.

4 Validation of Proposed Semi-analytical Solution

In this section, the proposed semi-analytical solution is validated by analyzing both laboratory and in-situ tests of rock-socketed shafts. For each test shaft, the complete stress–displacement curve was used to determine the “true” ultimate end-bearing capacity as detailed in the following sub-sections. The shaft dimensions and rock mass properties of each test are obtained from the original report or estimated from empirical values. Since the degree of disturbance from the construction to the rock mass is not known, three disturbance factor D values, $D=0.0$ (no disturbance), $D=0.5$ (medium disturbance), and $D=1.0$ (severe disturbance), are considered in the analyses.

4.1 Centrifuge Model Tests on Shafts in Pseudo-rock (Leung and Ko 1993)

Leung and Ko (1993) conducted centrifuge model tests on several shafts socketed in pseudo-rocks with unconfined compressive strengths below 12 MPa. The diameter and socket ratio of all test shafts were 26.6 mm and 3.15 mm. An acceleration of 40 g was applied in the tests, so the prototype diameter of the shafts should be 1.06 m. The detailed information about the two shafts that are analyzed here is summarized in Table 1. Since the pseudo-rock is gypsum cement-based, the m_i was determined as 8 according to Hoek and Brown (1997). The unconfined compressive strength σ_c was from the measured values in the original paper. The GSI values were taken from Gharsallaoui et al. (2020) who analyzed these shafts. The values of H_s , H_r , and B were simply from the original paper. In the table, the end-bearing capacity factor N_σ is defined as the ratio of the ultimate end-bearing capacity to the unconfined compressive strength of intact rock (q_u/σ_c). Figure 7 shows the measured base resistance-displacement curves for the two shafts. As can be seen, the shaft in the rock with $\sigma_c = 2.1$ MPa reached the “true” end-bearing capacity but the shaft in the rock with $\sigma_c = 6.7$ MPa did not when the test was stopped. By determining the measured q_u from the measured base resistance-displacement curve, the measured N_σ was determined as shown in Table 1. For the shaft in the rock with $\sigma_c = 6.7$ MPa, the measured N_σ is shown as 2.98+, where the “+” sign means the test shaft did not reach its “true” ultimate end-bearing capacity yet and the “true” N_σ could be larger.

4.2 Point Load Tests (Rehman and Broms 1971)

Rehman and Broms (1971) conducted point load tests to investigate the end-bearing capacity of shafts driven into various rocks. The tests were performed with steel dowels of 20 mm diameter on rock cubes encased in concrete. Two types

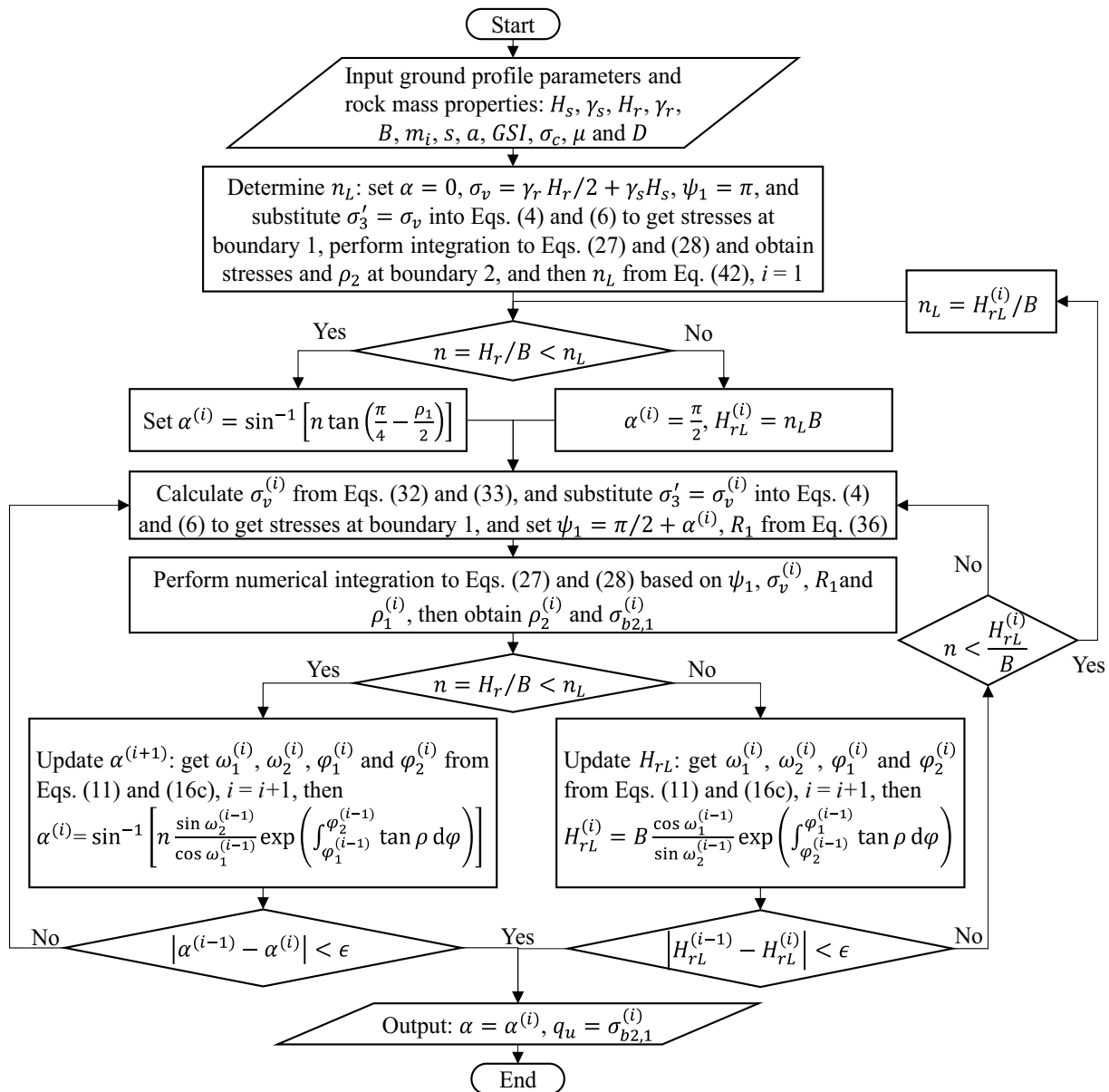


Fig. 6 Iterative solving procedure for proposed semi-analytical solution

of rocks, limestone and sandstone, were considered. Table 1 lists the rock properties, “shaft” dimensions, and measured N_σ . The σ_c was from the original paper, the m_i was determined following Hoek and Brown (1997), and the GSI was obtained from Gharsallaoui et al. (2020) who studied these tests. Figure 8 shows the measured load-tip settlement curves, from which the ultimate end-bearing capacity was determined and the measured N_σ was calculated.

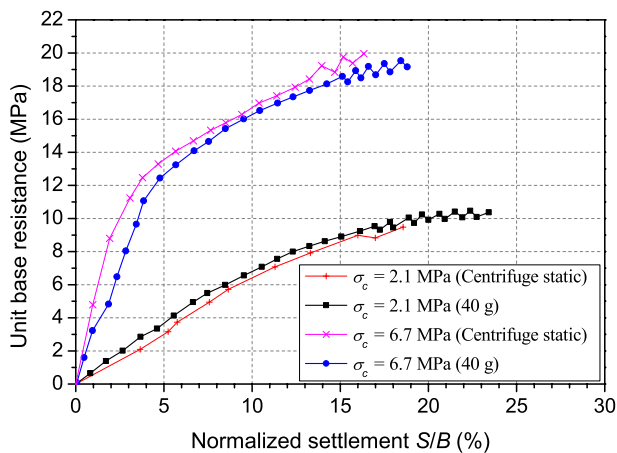
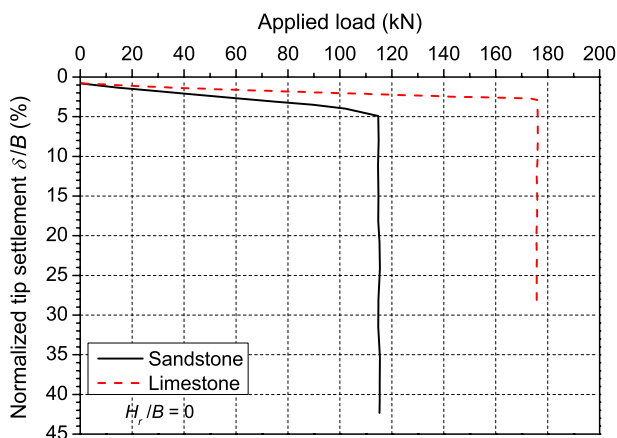
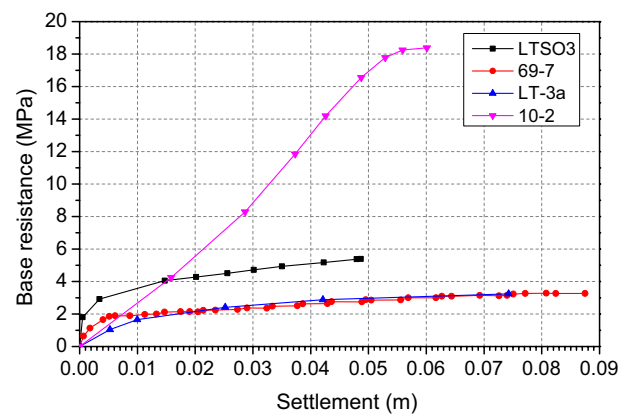
4.3 Osterberg Cell (O-Cell) Tests (Kim 2003)

Several O-cell tests were summarized by Kim (2003) to develop design guidelines for drilled shafts socketed in limestone in Florida. Four test shafts with full base resistance-settlement curves available for determining the ultimate end-bearing capacity are selected here for validation of the proposed solution. The ground profile, shaft dimensions and σ_c are obtained from Kim (2003) as shown in Table 1. The m_i was determined following Hoek and Brown (1997) and the GSI

Table 1 Summary of shaft tests and predictions from proposed solution

#	Resource	Rock type	m_i	σ_c (MPa)	GSI	H_s (m)	H_r (m)	B (m)	Measured N_σ^a	Predicted N_σ		
										$D=0$	$D=0.5$	$D=1$
1	Leung and Ko (1993)	Gypsum	8	2.1	73	0	3.0	1.06	4.90	8.54	6.50	3.97
2	Leung and Ko (1993)	Gypsum	8	6.7	73	0	3.3	1.06	2.98+	8.17	6.01	3.62
3	Rehman and Broms (1971)	Sandstone	17	70.5	75	0	0	0.02	5.21	8.20	6.11	3.51
4	Rehman and Broms (1971)	Limestone	10	124	75	0	0	0.02	4.50	5.48	4.09	2.37
5	Kim (2003)	Limestone	10	2.94	44	24.75	5.76	1.22	1.83+	8.63	5.05	1.93
6	Kim (2003)	Limestone	10	0.51	46	22.04	7.65	1.52	6.43	14.3	9.36	4.55
7	Kim (2003)	Limestone	10	4.12	38	0.08	8.33	1.83	0.79+	5.44	2.84	0.87
8	Kim (2003)	Limestone	10	15.03	37	12.50	6.30	1.22	1.22+	5.08	2.56	0.72

^aThe “+” sign means the shaft did not reach its ultimate end-bearing capacity at the end of the test. $\mu=0.25$ was adopted following Gercek (2007)


Fig. 7 Base resistance-settlement curves of centrifuge tests on shafts socketed in pseudo-rock (Leung and Ko 1993)

Fig. 8 Load-settlement curves of point load tests (Rehman and Broms 1971)

Fig. 9 Base resistance-settlement curves of O-Cell tests (Kim 2003)

was determined from the measured RQD using the following relation (Hoek et al. 2013):

$$GSI = 1.5 J_{cond_{89}} + RQD/2, \quad (43)$$

in which, $J_{cond_{89}}$ is the joint condition rating defined by Bieniawski (1989). Since $J_{cond_{89}}$ was not determined in the original report, an average value of 15 was assumed. Using the RQD values of 43, 47, 31, and 30 at sites LTISO3, 69-7, LT3a, and 10-2 (#5, 6, 7, and 8 in Table 1), respectively, the GSI values were calculated and are shown in Table 1. Again, using the base resistance-settlement curves as shown in Fig. 9, the ultimate end-bearing capacity was determined and the measured N_σ was calculated as shown in Table 1. Since shafts at sites LTISO3, LT3a, and 10-2 (#5, 7, and 8 in Table 1) did not reach the ultimate end-bearing capacity at the end of the test, the measured N_σ values are marked by the “+” sign.

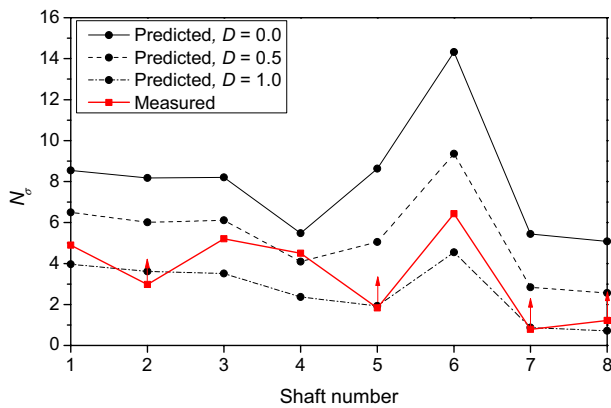


Fig. 10 Comparisons of measured and predicted N_σ values (upward arrow means the test shaft did not reach the “true” ultimate bearing capacity and the “true” N_σ should be higher)

4.4 Comparison of Measurements and Predictions

With the proposed semi-analytical solution, the N_σ of the eight (8) test shafts was determined and the predicted N_σ values are shown in Table 1. It is known that the construction process can cause disturbance to the rock mass and then affect the end-bearing capacity of rock-socketed shafts. According to AASHTO (2014), if blasting is used for shaft construction, a disturbance factor close to 1.0 should be considered. If rock coring is used to drill a hole, much less disturbance is expected and a disturbance factor of 0 may be considered. If a down-hole hammer is used to break up the rock, the disturbance is likely between these two extremes. Since no specific information is available about the construction method and the degree of disturbance for each test shaft, three different disturbance factor D values, 0.0 (no disturbance), 0.5 (medium disturbance) and 1.0 (severe disturbance), were used in the analyses. For a better comparison of the measured and predicted N_σ values, they are also plotted in Fig. 10. The points with an upward arrow in the figure mean that the test shaft did not reach the “true” ultimate end-bearing capacity at the end of test and the “true” measured N_σ should be higher. As can be seen, all the measured N_σ values (except shafts #2, #5 and #7) fall within the band formed by the predicted upper bound N_σ values at $D=0$

and the predicted lower bound N_σ values at $D=1$. Although the measured N_σ values for shafts #2, #5 and #7 are a bit below the predicted lower bound N_σ , the “true” measured N_σ should be higher because the “true” ultimate end-bearing capacity was not reached at the end of test for these three shafts. Therefore, it can be stated that the proposed semi-analytical solution provides good predictions of the ultimate end-bearing capacity of rock-socketed shafts. It can also be seen from the figure that the disturbance factor D has a great effect on the predicted N_σ and should be considered when determining the ultimate end-bearing capacity of rock-socketed shafts. If no information is available about the degree of disturbance from construction in practice, $D=0.5$ can be used in the analysis. It should be noted that the empirical determination of D should be further investigated by gathering more test shafts under different construction conditions.

5 Comparison with Existing Analytical and Numerical Results

The present solution is further compared with the analytical solution by Serrano et al. (2014) and the numerical simulation by Jafari et al. (2019) to explore the effects of 3D strength and 3D geometry. It is noted that the analytical solution of Serrano et al. (2014) was developed based on the 2D HB criterion and an empirical 3D factor proposed by DeBeer (1970). The numerical simulation by Jafari et al. (2019) was conducted as an axisymmetric problem by Plaxis 2D (Brinkgreve et al. 2019), with the rock mass governed by the 2D HB criterion and an unassociated flow rule, and the shaft modeled by an elastic model with a much higher stiffness than the rock mass.

The rock mass properties and shaft dimensions used in the comparative analyses are from Jafari et al. (2019) and summarized in Table 2. Figure 11 shows the end-bearing capacity factor N_σ versus the socket length ratio H_r/B at different m_i values based on the present solution, the analytical solution of Serrano et al. (2014) with the De Beer’s factor (De Beer 1970) to consider 3D geometry, and the numerical simulation by Jafari et al. (2019). As can be seen, N_σ increases with larger H_r/B , indicating the importance of socket length in increasing the end-bearing capacity

Table 2 Rock mass properties and shaft dimensions from Jafari et al. (2019)

#	m_i	σ_c (MPa)	GSI	D	E_{rm}/σ_c	$\psi_{\max}$ (°)	H_s (m)	B (m)
1	1	7	100	0	1000	8.7	0	1
2	5	7	100	0	1000	25.3	0	1
3	7.5	7	100	0	1000	30.5	0	1
4	10	7	100	0	1000	34.2	0	1

$\mu=0.25$ was adopted following Gercek (2007)

E_{rm} is the Young’s modulus of rock mass, $\psi_{\max}$ is the maximum dilation angle

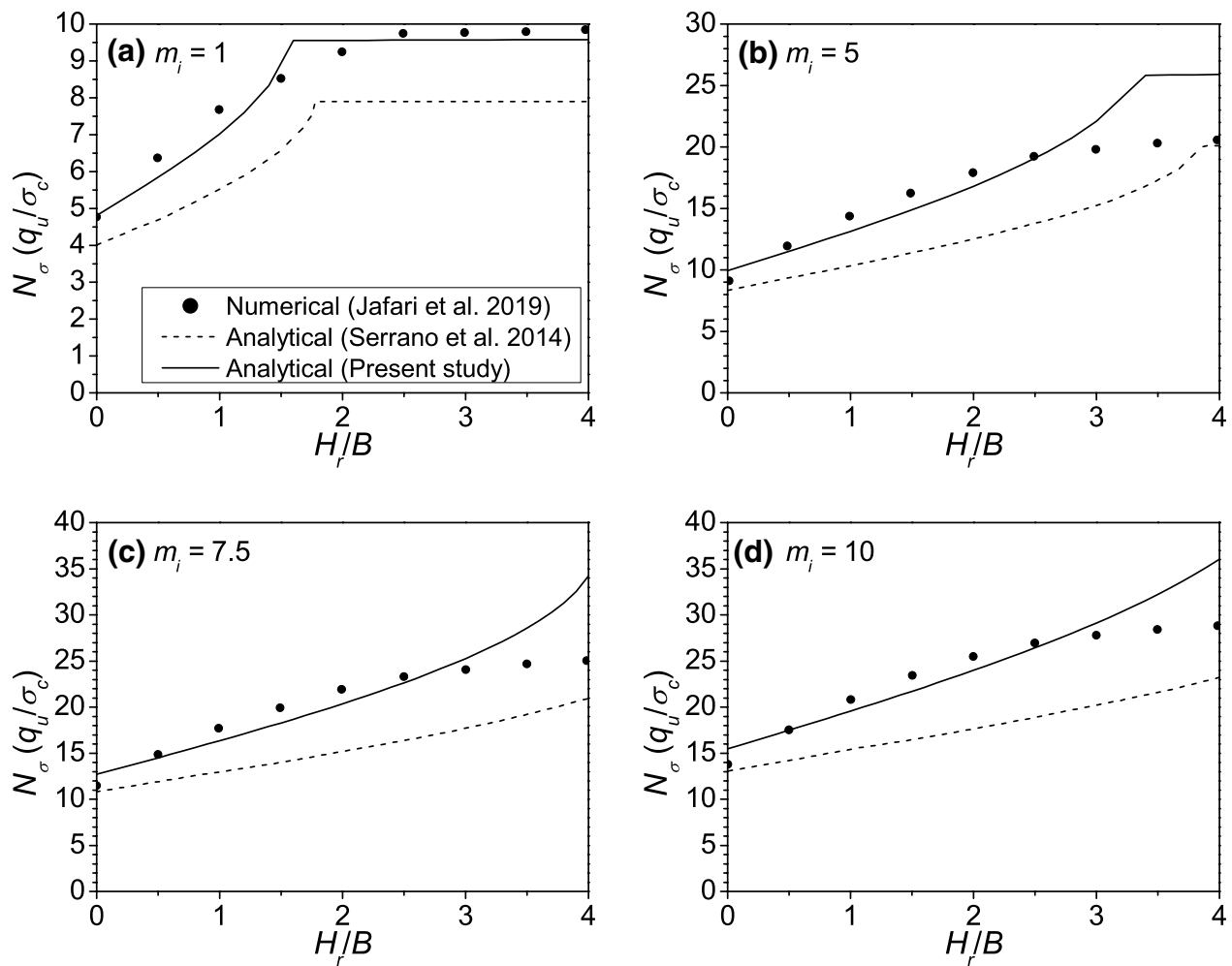


Fig. 11 Comparisons of proposed solution, 2D analytical solution with 3D effect factor by De Beer (1970) from Serrano et al. (2014), and numerical results from Jafari et al. (2019)

of rock-socketed shafts. The numerical results show that the increasing rate of N_σ decreases with H_r/B , while the increasing rates of N_σ predicted by the two solutions first increases with H_r/B and then suddenly decreases to zero. This is because the two solutions take the rock mass as a rigid-plastic material and ignore the effects of settlement. The proposed solution makes slightly higher estimations of the end-bearing capacity of rock-socketed shafts than the numerical simulation when $\frac{H_r}{B} > 2.5$ with $m_i = 5, 7.5$ and 10, because the present solution considers the 3D strength, but the numerical simulation does not. In contrast, the predictions of the analytical solution by Serrano et al. (2014)

based on the 2D HB criterion and the De Beer's factor (De Beer 1970) are smaller than both the numerical results and the present solution. This is because the solution of Serrano et al. (2014) only empirically considers the 3D geometry effect and ignores the 3D strength of rock mass. This indicates that the empirical factor proposed by De Beer (1970) only partially considers 3D effects on the end-bearing capacity of rock-socketed shafts. Therefore, it is important to consider both 3D strength and 3D geometry in the analysis of rock-socketed shafts.

Table 3 Rock mass properties and shaft dimensions for parametric analyses

#	m_i	σ_c (MPa)	GSI	H_s (m)	H_r (m)	B (m)	μ^a
a	5	5	75	8	1	1	0.25
b	20	5	75	8	1	1	0.25

^a $\mu = 0.25$ was adopted following Gercek (2007)

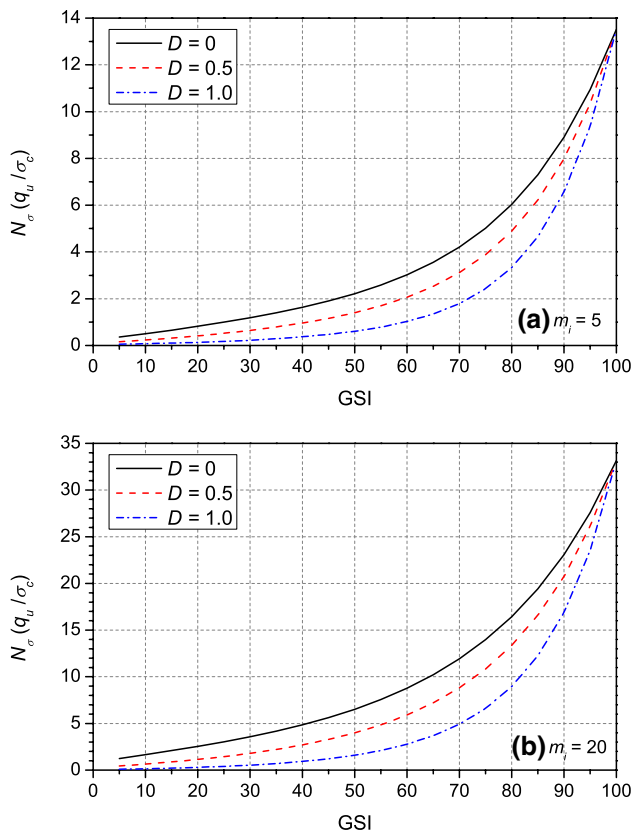


Fig. 12 Effect of GSI on end-bearing capacity factor N_σ : **a** $m_i = 5$; and **b** $m_i = 20$

6 Parametric Analyses

In this section, extensive parametric analyses are performed using the proposed solution to investigate the effects of rock mass properties (GSI and σ_c) and shaft dimensions (H_r and H_s) on the end-bearing capacity of rock-socketed shafts. For the analyses, two m_i values (5 and 20) and three different D values (0.0, 0.5, and 1.0) are considered. The basic parameters for the rock masses and shaft dimensions are provided in Table 3, and the parametric analyses are carried out by changing one of those parameters (GSI, σ_c , H_r and H_s) at a time.

6.1 Effect of GSI

Figure 12 plots the end-bearing capacity factor N_σ versus GSI with $m_i = 5$ and 20, respectively. As expected, at the same D and GSI values, the N_σ at $m_i = 20$ is much greater than that at $m_i = 5$. At both m_i values and the various D values, the N_σ increases with higher GSI and the increasing rate becomes larger when GSI is higher. It is noted that the effect of the disturbance factor D is negligible when GSI is close to 100. This is because when GSI is close to 100, the

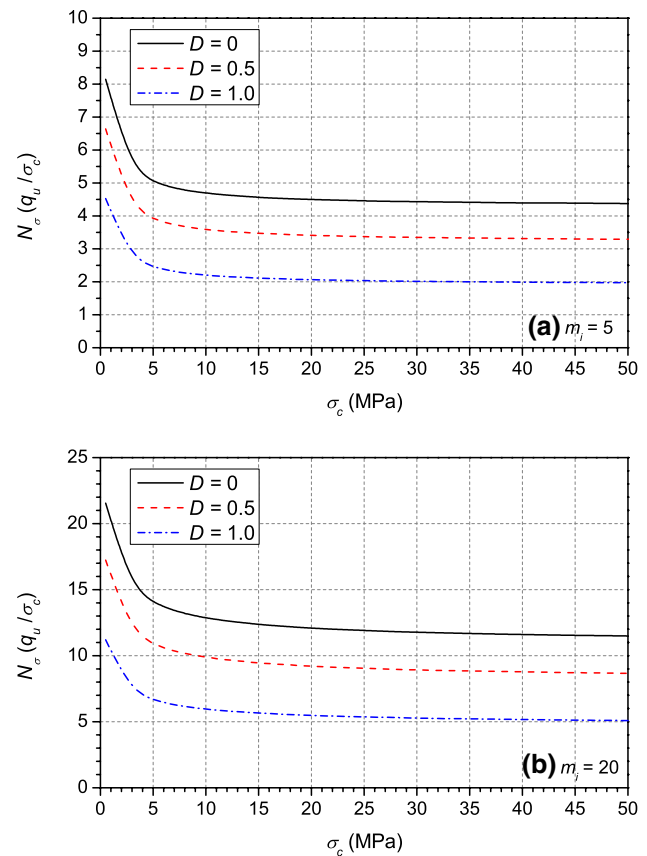


Fig. 13 Effect of σ_c on end-bearing capacity factor N_σ : **a** $m_i = 5$; and **b** $m_i = 20$

disturbance factor D barely affects the rock mass properties as indicated by Eqs. (5e)–(5g). However, the effect of the disturbance factor D is significant when $GSI < 90$, with the N_σ decreasing significantly when the D goes from 0 to 0.5 then to 1.0. So, for shafts socketed in a rock mass with $GSI < 90$, it is important to use proper construction methods and reduce the disturbance on the rock mass due to the construction process.

6.2 Effect of σ_c

Figure 13 shows the effect of σ_c on the end-bearing capacity factor N_σ with $m_i = 5$ and 20, respectively. At the same D and σ_c values, the N_σ at $m_i = 20$ is much greater than that at $m_i = 5$. At both m_i values and the various D values, the N_σ decreases with higher σ_c and the decreasing rate becomes smaller when σ_c gets larger. When σ_c is larger than 30 MPa, the N_σ essentially does not change with σ_c anymore. It can also be seen that the disturbance factor D has a great effect on the N_σ at all σ_c values. So, it is important to use proper construction methods and reduce the disturbance on the rock mass due to the construction process.

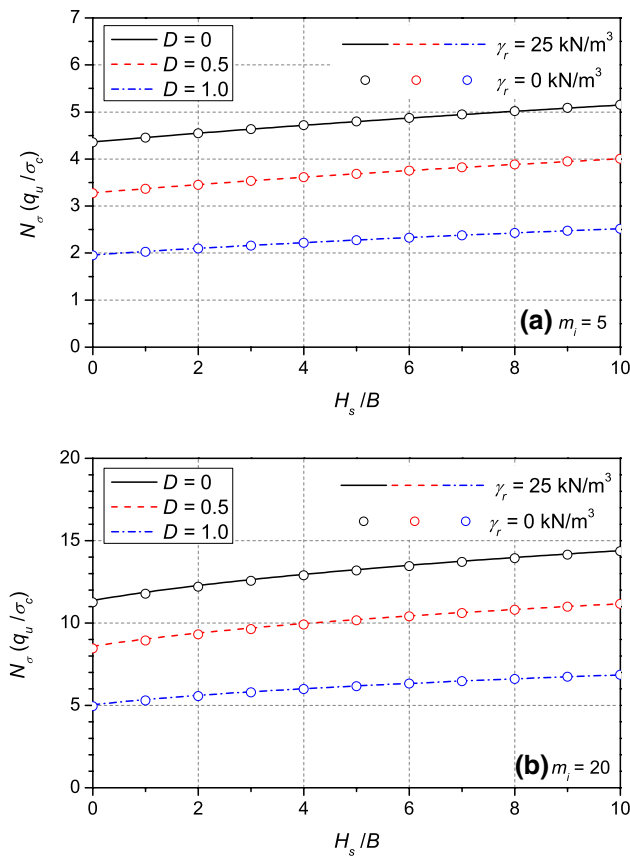


Fig. 14 Effect of H_s/B on end-bearing capacity factor N_σ : **a** $m_i = 5$; and **b** $m_i = 20$

6.3 Effect of H_s

Figure 14 plots the N_σ versus H_s/B with $m_i = 5$ and 20, respectively. Both $\gamma_r = 0 \text{ kN/m}^3$ and 25 kN/m^3 in Eq. (20) are considered to investigate the effect of the self-weight of rock mass on the ultimate end-bearing capacity. The results with $\gamma_r = 0 \text{ kN/m}^3$ are plotted as empty dots while those with $\gamma_r = 25 \text{ kN/m}^3$ are plotted as lines. At the same D and H_s/B values, the N_σ at $m_i = 20$ is much greater than that at $m_i = 5$. H_s mainly affects the overburden pressure, as indicated by Eqs. (32) and (33). At both m_i values and the various D values, increasing H_s from 0 to $10B$ only slightly increases the N_σ . Therefore, the overburden pressure caused by H_s only has a minor effect on the ultimate end-bearing capacity of rock-socketed shafts. Again, the disturbance factor D has a great effect on the N_σ at all H_s/B values. Therefore, it is important to use the right construction method to minimize the disturbance to the rock mass from the construction process. As anticipated, the results with $\gamma_r = 0 \text{ kN/m}^3$ and 25 kN/m^3 show very little difference ($< 1\%$), meaning that the self-weight of rock mass has essentially no effect on the ultimate end-bearing capacity of a rock-socketed shaft.

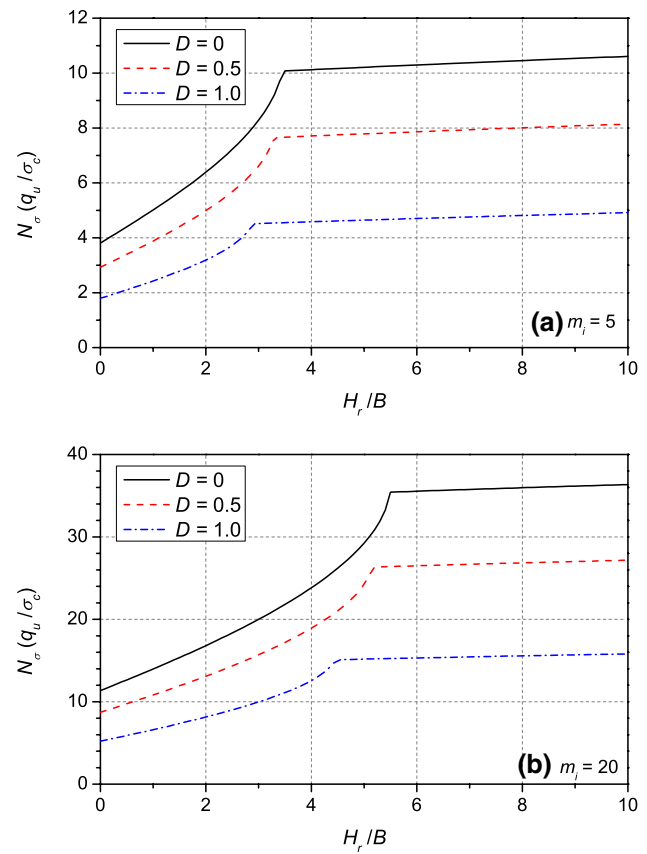


Fig. 15 Effect of H_r/B on end-bearing capacity factor N_σ : **a** $m_i = 5$; and **b** $m_i = 20$

6.4 Effect of H_r

Figure 15 plots the N_σ versus H_r/B with $m_i = 5$ and 20, respectively. Since the rock mass is taken as a rigid-plastic material in this study, a kink can be clearly observed in each curve due to ignorance of elastic displacement. At the same D and H_r/B values, the N_σ at $m_i = 20$ is much greater than that at $m_i = 5$. At both $m_i = 5$ and 20, the N_σ first increases dramatically with H_r/B , which corresponds to the inclination angle α in Fig. 1 increasing from 0° to 90° . Then, after α reaches 90° , the failure surface closes on the shaft side, and the increase of H_r/B will no longer change the shape of the failure surface, leading to the kink on the $N_\sigma-H_r/B$ curve. After the kink, the increase of H_r/B only causes the increase of overburden pressure as indicated by Eq. (33), and thus only small increases of N_σ . The $H_r/B (=n)$ at the kink point is the critical socket ratio n_L . The n_L increases with higher m_i and lower D because a rock mass with higher m_i and lower D tends to have a larger instant friction angle ρ and thus a larger log-spiral failure surface. Since N_σ only slightly increases with H_r/B after H_r/B reaches n_L , to maximize the end-bearing resistance without using too long a socket depth, a H_r/B equal

to the critical socket ratio n_L (which is between 3 and 5.5 for m_i between 5 and 20) can be used.

7 Conclusion

A new semi-analytical solution is proposed for determining the ultimate end-bearing capacity of rock-socketed shafts with the consideration of both 3D strength and 3D geometry. The proposed solution is validated using it to analyze eight test shafts. The proposed solution is also compared with an analytical solution based on 2D HB criterion and an empirical factor proposed by De Beer (1970) for considering the 3D effect, and a numerical simulation based on 2D HB criterion and 3D geometry to investigate the effects of 3D strength and 3D geometry. Finally, extensive parametric analyses are carried out to study the effects of rock mass properties, shaft dimensions, and the disturbance factor with the proposed solution. The major conclusions of the present study can be drawn as follows:

1. The proposed solution considering both 3D strength and 3D geometry makes reasonable predictions of the ultimate end-bearing capacity q_u of rock-socketed shafts. The De Beer's factor only partially considers the 3D effect.
2. It is important to consider the disturbance to the rock mass from construction when evaluating the ultimate end-bearing capacity q_u of rock-socketed shafts. If no information is available about the degree of disturbance from construction, a disturbance factor $D=0.5$ can be used in the analysis.
3. Ignoring 3D strength and 3D geometry could significantly underestimate the ultimate end-bearing capacity q_u of rock-socketed shafts.
4. Besides the σ_c , the m_i , GSI and H_r can also significantly affect the ultimate end-bearing capacity q_u of rock-socketed shafts and thus should be considered when determining the q_u .
5. The disturbance factor D significantly affects the ultimate end-bearing capacity q_u of rock-socketed shafts. To ensure high q_u , it is important to use the right construction method to minimize the disturbance to the rock mass from the construction process.

Appendix

A.1 Details of Eqs. (24a–24e)

$$\frac{\partial f}{\partial \sigma_i^*} = \frac{\partial f}{\partial I_1^*} + \frac{\partial f}{\partial q} \frac{3\sigma_i^* - I_1^*}{2q} + \frac{\partial f}{\partial I_3^*} \frac{I_3^*}{\sigma_i^*}; i = 1, 2, 3, \quad (44a)$$

$$\frac{\partial f}{\partial I_1^*} = \left[\left(\frac{q}{\sigma_c} \right)^{2/a-2} + m_b \left(\frac{q}{\sigma_c} \right)^{1/a-1} \right] \frac{3I_1^{*2} - q^2}{3}, \quad (44b)$$

$$\begin{aligned} \frac{\partial f}{\partial q} = & - \left[\left(\frac{q}{\sigma_c} \right)^{2/a-2} + m_b \left(\frac{q}{\sigma_c} \right)^{1/a-1} \right] \frac{2qI_1^*}{3} \\ & + \left[\frac{2-2a}{a} \left(\frac{q}{\sigma_c} \right)^{2/a-2} + m_b \frac{1-a}{a} \left(\frac{q}{\sigma_c} \right)^{1/a-1} \right] \left(\frac{I_1^{*3} - I_1^* q^2}{3q} - \frac{9I_3^*}{q} \right), \end{aligned} \quad (44c)$$

$$\frac{\partial f}{\partial I_3^*} = -9 \left[\left(\frac{q}{\sigma_c} \right)^{2/a-2} + m_b \left(\frac{q}{\sigma_c} \right)^{1/a-1} \right] - 2m_b^2. \quad (44d)$$

A.2 Details of Eq. (26b)

$$A_{13} = - \frac{\frac{\partial f}{\partial \sigma_1^*} + \mu \frac{\partial f}{\partial \sigma_2^*}}{\frac{\partial f}{\partial \sigma_3^*} + \mu \frac{\partial f}{\partial \sigma_2^*}}, \quad (45a)$$

$$\frac{dA_{13}}{d\sigma_1} = - \frac{\frac{\partial^2 f}{\partial \sigma_1^* \partial \sigma_3^*} + \mu \frac{\partial^2 f}{\partial \sigma_1^* \partial \sigma_2^*}}{\frac{\partial f}{\partial \sigma_3^*} + \mu \frac{\partial f}{\partial \sigma_2^*}} + \frac{\frac{\partial f}{\partial \sigma_1^*} + \mu \frac{\partial f}{\partial \sigma_2^*}}{\left(\frac{\partial f}{\partial \sigma_3^*} + \mu \frac{\partial f}{\partial \sigma_2^*} \right)^2} \left(\frac{\partial^2 f}{\partial \sigma_1^* \partial \sigma_3^*} + \mu \frac{\partial^2 f}{\partial \sigma_1^* \partial \sigma_2^*} \right), \quad (45b)$$

$$\frac{dA_{13}}{d\sigma_2} = - \frac{\frac{\partial^2 f}{\partial \sigma_1^* \partial \sigma_2^*} + \mu \frac{\partial^2 f}{\partial \sigma_2^* \partial \sigma_3^*}}{\frac{\partial f}{\partial \sigma_3^*} + \mu \frac{\partial f}{\partial \sigma_2^*}} + \frac{\frac{\partial f}{\partial \sigma_1^*} + \mu \frac{\partial f}{\partial \sigma_2^*}}{\left(\frac{\partial f}{\partial \sigma_3^*} + \mu \frac{\partial f}{\partial \sigma_2^*} \right)^2} \left(\frac{\partial^2 f}{\partial \sigma_2^* \partial \sigma_3^*} + \mu \frac{\partial^2 f}{\partial \sigma_2^* \partial \sigma_1^*} \right), \quad (45c)$$

$$\frac{dA_{13}}{d\sigma_3} = - \frac{\frac{\partial^2 f}{\partial \sigma_1^* \partial \sigma_3^*} + \mu \frac{\partial^2 f}{\partial \sigma_2^* \partial \sigma_3^*}}{\frac{\partial f}{\partial \sigma_3^*} + \mu \frac{\partial f}{\partial \sigma_2^*}} + \frac{\frac{\partial f}{\partial \sigma_1^*} + \mu \frac{\partial f}{\partial \sigma_2^*}}{\left(\frac{\partial f}{\partial \sigma_3^*} + \mu \frac{\partial f}{\partial \sigma_2^*} \right)^2} \left(\frac{\partial^2 f}{\partial \sigma_3^* \partial \sigma_1^*} + \mu \frac{\partial^2 f}{\partial \sigma_3^* \partial \sigma_2^*} \right), \quad (45d)$$

where

$$\begin{aligned} \frac{\partial^2 f}{\partial \sigma_i^{*2}} = & \frac{\partial^2 f}{\partial I_1^{*2}} + \frac{\partial^2 f}{\partial I_1^* \partial q} \frac{3\sigma_i^* - I_1^*}{q} + \frac{\partial f}{\partial q} \frac{1}{q} \left[1 - \left(\frac{3\sigma_i^* - I_1^*}{2q} \right)^2 \right] \\ & + \frac{\partial^2 f}{\partial q^2} \left(\frac{3\sigma_i^* - I_1^*}{2q} \right)^2 + \frac{I_3^*}{q} \frac{3\sigma_i^* - I_1^*}{\sigma_i^*} \frac{\partial^2 f}{\partial q \partial I_3^*}; i = 1, 2, 3, \end{aligned} \quad (46a)$$

$$\begin{aligned} \frac{\partial^2 f}{\partial \sigma_i^* \partial \sigma_j^*} = & \frac{\partial^2 f}{\partial I_1^{*2}} + \frac{\partial^2 f}{\partial I_1^* \partial q} \frac{3\sigma_i^* + 3\sigma_j^* - 2I_1^*}{2q} \\ & - \frac{\partial f}{\partial q} \frac{1}{q} \left[\frac{1}{2} + \left(\frac{3\sigma_i^* - I_1^*}{2q} \right) \left(\frac{3\sigma_j^* - I_1^*}{2q} \right) \right] \\ & + \frac{\partial^2 f}{\partial q^2} \left(\frac{3\sigma_i^* - I_1^*}{2q} \right) \left(\frac{3\sigma_j^* - I_1^*}{2q} \right) \\ & + \frac{\partial^2 f}{\partial q \partial I_3^*} \frac{I_3^*}{2q} \left(\frac{3\sigma_i^* - I_1^*}{\sigma_j^*} + \frac{3\sigma_j^* - I_1^*}{\sigma_i^*} \right) \\ & + \frac{\partial f}{\partial I_3^*} \frac{I_3^*}{\sigma_i^* \sigma_j^*}; i \neq j, \end{aligned} \quad (46b)$$

$$\frac{\partial^2 f}{\partial I_1^2} = 2I_1^* \left[\left(\frac{q}{\sigma_c} \right)^{\frac{2}{a}-2} + m_b \left(\frac{q}{\sigma_c} \right)^{\frac{1}{a}-1} \right], \quad (46c)$$

$$\begin{aligned} \frac{\partial^2 f}{\partial I_1^* \partial q} &= -\frac{2q}{3} \left[\left(\frac{q}{\sigma_c} \right)^{\frac{2}{a}-2} + m_b \left(\frac{q}{\sigma_c} \right)^{\frac{1}{a}-1} \right] \\ &+ \left[\frac{2-2a}{a} \left(\frac{q}{\sigma_c} \right)^{\frac{2}{a}-3} + m_b \frac{1-a}{a} \left(\frac{q}{\sigma_c} \right)^{\frac{1}{a}-2} \right] \left(\frac{3I_1^{*2} - q^2}{3\sigma_c} \right), \end{aligned} \quad (46d)$$

$$\begin{aligned} \frac{\partial^2 f}{\partial q^2} &= -\frac{2I_1^*}{3} \left[\left(\frac{q}{\sigma_c} \right)^{\frac{2}{a}-2} + m_b \left(\frac{q}{\sigma_c} \right)^{\frac{1}{a}-1} \right] \\ &- \frac{4I_1^*}{3} \left[\frac{2-2a}{a} \left(\frac{q}{\sigma_c} \right)^{\frac{2}{a}-2} + m_b \frac{1-a}{a} \left(\frac{q}{\sigma_c} \right)^{\frac{1}{a}-1} \right] \\ &+ \left[\frac{2-2a}{a} \frac{2-3a}{a} \left(\frac{q}{\sigma_c} \right)^{2/a-4} + m_b \frac{1-a}{a} \frac{1-2a}{a} \left(\frac{q}{\sigma_c} \right)^{1/a-3} \right] \\ &\times \left(\frac{I_1^{*3} - I_1^* q^2}{3\sigma_c^2} - \frac{9I_3^*}{\sigma_c^2} \right), \end{aligned} \quad (46e)$$

$$\frac{\partial^2 f}{\partial I_3^* \partial q} = -\frac{9}{\sigma_c} \left[\frac{2-2a}{a} \left(\frac{q}{\sigma_c} \right)^{\frac{2}{a}-3} + m_b \frac{1-a}{a} \left(\frac{q}{\sigma_c} \right)^{\frac{1}{a}-2} \right]. \quad (46f)$$

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